

Enhancing Accessibility and Equity in MIT's Transportation Systems

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1 Introduction and Background

This project aims to improve the accessibility and equity of transportation options for the MIT community, with a particular focus on low-income students and staff living farther from campus.

Reliable, affordable, and safe mobility is essential for student well-being, campus sustainability, and community inclusion. Despite MIT's robust commuter benefits for faculty and staff, many students and non-faculty staff continue to face significant barriers to affordable, sustainable, and convenient mobility.

MIT's transportation landscape is characterized by a reliance on personal vehicles and limited public transit options for some community members. Key obstacles include the high cost and limited convenience of public transit for occasional users, insufficient access to safe and affordable biking (especially e-bikes), and persistent inequities in parking costs for staff who cannot rely on public transit. Many students report skipping off-campus activities or walking long distances with heavy groceries to avoid MBTA fares, while staff who live far from MBTA lines and arrive before 2:30pm must pay substantial daily or monthly parking fees.

This project builds on the idea of expanding access to current systems rather than creating new ones, targeting both physical and financial barriers to sustainable transportation. Our approach is grounded in data-driven recommendations and campus community feedback, with a specific emphasis on prioritizing the needs of low-income students and staff. Drawing on two extensive campus-wide surveys ($n > 160$ and $n > 180$), focus groups, and direct testimonials, we identify and address transportation equity at MIT through three key pillars:

- **Public Transit:** Addressing the high cost and limited convenience of MBTA service for students, and the lack of a daytime MIT shuttle to Central Square, which limits access to affordable groceries and the Red Line.
- **Biking:** Improving Bluebikes e-bike access and affordability, and reducing safety and intimidation barriers for new bikers and those lacking urban biking experience.
- **Parking Equity for Staff:** Tackling the persistent inequity where custodial and facilities staff pay the same parking rates as nearby faculty, despite having fewer alternatives, especially for those who must arrive before free parking hours.

By leveraging existing infrastructure and exploring targeted new initiatives, this project seeks to reduce reliance on personal vehicles, enhance mobility, and contribute to MIT's sustainability and equity goals.

2 Objectives

2.1 Enhancing Public Transit Accessibility for MIT Community

Problem Statement

The stretch of Massachusetts Avenue connecting Harvard Square, Central Square, MIT, and Back Bay is a critical route for MIT students. Harvard offers numerous restaurants in addition to being a destination for cross-registrees; while Central provides our closest and most diverse selection of grocery store options, as well as the closest MBTA Red Line station; and Back Bay houses many FSILGs and additional grocery options like Trader Joe's and Star Market.

While MIT Tech Shuttles connect campus to Back Bay, there is no daytime shuttle service to Central Square, despite it being home to essential services like Daily Table (a lifeline for many low-income students). On-campus grocery options like Concord Market remain unaffordable for many, making reliable transportation to Central Square crucial for food security and student well-being.

MIT offers subsidized MBTA passes, but these are still incredibly expensive and only cost-effective for daily commuters. This creates a significant barrier for students who need to make infrequent trips for groceries, medical care, or social activities. Survey data from over 160 respondents shows that 90% of undergraduate students have opted out of trips or carried heavy cargo (e.g., groceries) for long distances to avoid transit fares. Many shared experiences of walking 20–30 minutes with heavy groceries, sometimes in poor weather, or simply skipping events and outings due to cost. Representative student comments include:

“I walk a mile in the rain with heavy groceries because I can't justify \$2.40 for the T.”

“Carrying a gallon of milk and a full bag of groceries is fine until you have to carry it for half an hour walking back home from the store... Many such instances unfortunately.”

“I have frequently had to skip meals during the day because there are few good options for me to inexpensively commute home and access my groceries.”

“Bought a ton of groceries at market basket in somerville and walked to a union square bus stop then took a bus to central square but the next 1 bus was in like 30 min, so I carried all my groceries from central to macgregor. Many regrets, my back hurt for a couple of days.”

“Sometimes since leaving campus can be such a hassle [it's] much easier to just stay in which takes a toll on mental health for sure.”

Destinations students wish to access more easily include Trader Joe’s, Daily Table, Target, medical care, pharmacies, and social activities across Boston and Cambridge. Many noted that even short trips can become “an event” due to the lack of direct, affordable transit options.

Institutional Updates and Opportunities

- **M2 Shuttle Partnership in Progress:**

Currently, the Longwood Collective LMA M2 Shuttle requires a valid Harvard ID for access. MIT students have sometimes used Harvard Library cards to board, although these are not officially recognized as valid IDs.¹

MIT’s shuttles, excluding on-demand saferide services, are open to the public without ID checks yet rarely become overwhelmed. This openness suggests that a reciprocal arrangement could enhance collaboration and mutual benefit between MIT and Harvard, allowing both institutions to share resources efficiently.

MIT Parking and Transportation is actively negotiating with Harvard for MIT student access to the Longwood Collective M2 Shuttle (Harvard ↔ Central ↔ MIT ↔ Longwood Medical Area). This route directly addresses gaps in access to affordable groceries, cross-registration, and medical care. As of May 2025, both institutions are gathering data and negotiating terms, with a potential launch as early as Fall 2025.

- **Undergraduate Association (UA) Public Affairs Advocacy:**

The Undergraduate Association’s Public Affairs Committee (UAPA) is once again considering a pilot for free MBTA passes for MIT undergraduates, potentially through the MBTA University Pass program. This renewed interest is inspired by Tufts University’s recent University Pass pilot, which provides SMFA undergraduates with unlimited, free subway and bus access in exchange for a flat monthly fee paid by Tufts.² This program is open to other colleges and, notably, allows targeting specific student groups, such as low-income students.³

I will note that, from what I understand, a previous proposal to grant all MIT undergraduates free MBTA passes—by folding the cost into tuition, which would have been covered by financial aid for low-income students—was previously tabled but ultimately blocked by the UA. At the time, the UA had a high proportion of higher-income student representatives who expressed concerns about raising tuition for all students, even if the change would not have affected those on financial aid. This history highlights the need for a targeted, needs-based pilot that ensures support reaches those most affected by transit costs while still avoiding unintended financial burdens on students who can least afford them.

¹<https://www.longwoodcollective.org/about-shuttles>

²<https://now.tufts.edu/2024/09/03/new-pilot-program-offers-free-mbta-passes-smfa-degree-candidates>

³<https://www.mbta.com/news/2024-06-24/mbta-and-tufts-university-partner-new-university-pass-pilot-benefitting-hundreds>

Recommendations

1. **Support and Monitor the M2 Shuttle Partnership:** Continue to work closely with MIT Parking & Transportation and Harvard to ensure that the proposed M2 Shuttle access for MIT students is implemented effectively. Advocate for a launch timeline that addresses critical needs such as grocery access, cross-registration, and medical appointments. Track progress and collect user feedback after rollout to inform future improvements.
2. **Pilot a Targeted Free MBTA Pass Program:** Collaborate with UAPA and administration to pilot a free MBTA pass for low-income undergraduates (rather than the entire undergraduate student body) modeled on the MBTA/Tufts University Pass program. Even a pilot limited to local bus routes (if full subway access is not financially feasible) would significantly enhance mobility and reduce financial barriers.⁴ This targeted approach avoids tuition increases for all students while ensuring that support reaches those most affected by transit costs. Use data from this pilot to assess impact and build the case for broader implementation.
3. **Improve Communication and Outreach:** Increase awareness of existing MBTA subsidies and shuttle services. Provide clear, accessible information about all available transit options, including eligibility and application processes.
4. **Explore Flexible Transit Subsidies:** Advocate for more flexible, needs-based transit subsidies that better serve occasional users and those with irregular schedules.
5. **Other potential solutions** involve expanding MIT shuttle services by adding a Central Square stop to an existing route or creating a new route between 77 Mass Ave., Random Hall, and Central Square. However, these approaches require additional resources and infrastructure, which may be less feasible than leveraging existing systems.

Enhancing public transit accessibility is essential for reducing financial and logistical barriers to food security, healthcare, and community engagement at MIT. By supporting institutional partnerships, piloting targeted subsidies, and amplifying student voices, MIT can make meaningful progress toward a more equitable and sustainable campus transportation system.

⁴<https://news.mit.edu/2019/public-transportation-discount-affect-low-income-0619>

2.2 Enhancing Bluebikes Accessibility for MIT Community

Problem Statement

Biking is a sustainable, affordable, and healthy transportation option, but significant barriers prevent many MIT students and staff from taking advantage of it. These include:

- **Limited E-Bike Availability:** Bluebikes e-bikes are rarely available at MIT-area stations, and the hilly topography (e.g., Mass Ave uphill in both directions) and harsh Boston winters make e-bikes especially valuable for longer or more difficult trips.
- **Financial Barriers:** Even with MIT’s subsidized annual Bluebikes membership (\$50.75, regularly \$133.50), e-bike usage incurs an additional per-minute fee. This cost deters low-income students and staff from using e-bikes, despite their potential benefits. Staff and faculty who opt into the MIT bike benefit cannot also receive the free MBTA pass, forcing a difficult choice for those who live far from transit lines.
- **Safety, Intimidation, and Lack of Experience:** Many students feel unsafe biking in Cambridge/Boston traffic, and a notable minority do not know how to ride a bike at all. There is a lack of structured support for learning to bike or for building confidence in urban riding.
- **Under-Promotion of Subsidies and Resources:** MIT’s existing Bluebikes subsidy and other biking resources are not widely known or utilized.

Survey Insights ($n = 185$)

- **Comfort Biking in Cambridge/Boston Traffic:**
 - 18.4% feel very comfortable
 - 19.5% have never biked in city traffic
 - 38.4% somewhat comfortable, 16.2% not very comfortable, 7.6% not at all comfortable
- **Barriers and Supports Requested:**
 - 86.9% want better/protected bike lanes and infrastructure
 - 61.7% want maps/resources on safe biking routes
 - 50.8% want more secure bike parking
 - 45.9% want safety/training workshops
 - 42.6% want access to affordable/subsidized helmets and safety gear
 - 33.3% want group rides or mentorship programs
 - Notable write-ins: *learning hand signals, funds for repairs/maintenance*
 “The main thing I fear is that people who are low income [don’t] prioritize maintenance repairs (brake pads) due to the cost or getting second hand bikes, so having some funds for people would I think make people more confident”

- **Interest in MIT PE Biking Classes:**

- 69%+ would take a class if it covers street-biking safety and urban riding, group rides, or practical city biking experience
- Strong demand for absolute beginner (even if only via semesterly “learn-to-bike” events) and intermediate/advanced options
- Student voices:
 - “Please do this!!!!!! I’m desperate to learn how to ride a bike”
 - “A biking PE class would be awesome! I know many people who want to learn but there are too many barriers, such as dangers of city biking, and also inability to acquire a (decent) bike for cheap.”

The recent federal grant supporting Bluebikes expansion presents an opportunity to advocate for more stations in urban areas like Cambridge.^{5,6}

Recommendations

1. **Advocate for Increased E-Bike Allocation:** Use survey data to request that Bluebikes prioritize e-bike distribution at MIT-area stations, especially during peak commuting hours and winter months.
2. **Expand Financial Support:** Further subsidize or eliminate the per-minute e-bike fee for low-income students and staff, and consider making annual Bluebikes memberships free for those in need.
3. **Boost Awareness and Communication:** Launch a targeted campaign to promote the Bluebikes subsidy and other biking resources. Promote the benefits of biking over ride-hailing services or personal vehicles to reduce carbon emissions and enhance sustainability.⁷

⁵<https://www.mapc.org/news/fed-grant-to-fund-congestion-relief-solutions/>

⁶<https://mass.streetsblog.org/2024/10/22/federal-funds-will-let-bluebikes-system-add-electrified-stations-for-e-bikes>

⁷<https://marcommnews.com/city-of-boston-launches-new-get-your-pass-in-gear/>

4. **Build Biking Confidence and Skills:** Develop programs to increase comfort and confidence in biking:⁸

- **Workshops and Training:** Offer workshops specifically for new bikers or women, focusing on street safety and bike etiquette.
- **Partnerships:** Collaborate with organizations like Bikes Not Bombs to provide basic bike maintenance training, repair support, and access to affordable bikes.
- **MIT Resources:**
 - Partner with MIT Physical Education to offer bike safety workshops, ensuring students are equipped with necessary skills for safe commuting.
 - Provide free or affordable helmets and safety gear for low-income students with subsidized Bluebikes memberships.

5. **Policy Review for Staff/Faculty Benefits:** Re-examine the “either/or” policy between the bike benefit and MBTA pass for staff/faculty, seeking a more flexible approach for those with hybrid commutes.

Improving biking access, affordability, and safety at MIT will reduce transportation inequities, promote sustainability, and empower more students and staff to choose active, healthy, and low-carbon mobility options. Student demand for biking education and support is clear; MIT should act on these findings to make biking a truly accessible and attractive option for all.

⁸<https://betterbikeshare.org/2023/05/23/better-bike-share-partnership-awards-100k-in-grants/>

2.3 Addressing Faculty and Staff Transportation Needs

Problem Statement

While the primary focus of this project is on student transportation equity,⁹ faculty and staff commuting patterns also present significant opportunities for improvement. Post-pandemic remote work policies have reduced overall parking demand, but persistent inequities remain for staff—especially custodial and facilities workers—living far from campus or MBTA lines due to high housing costs. This results in a greater reliance on personal vehicles for commuting, particularly for those working non-traditional shifts, when public transit options are limited.

Although all benefits-eligible MIT faculty and staff receive free, unlimited MBTA subway and bus access via the Access MIT program, this benefit is not universally useful. Staff who live in areas not served by the MBTA, or who must arrive before 2:30pm (when parking is still paid), face daily or monthly parking costs (\$11/day or \$174/month) that can represent a significant financial burden. In contrast, faculty who live close to campus or along MBTA lines may choose not to use their free transit benefit and still pay the same parking rates as staff with fewer alternatives.

Recommendations

1. **Implement a Sliding Scale for Parking Fees:** Develop a parking fee structure based on income and role, ensuring that custodial and facilities staff who must drive and arrive before 2:30pm pay less than higher-salaried faculty or administrators. This approach recognizes the disparate financial burdens and limited commuting alternatives faced by non-administrative staff.
2. **Review and Adjust Commuter Benefits Policies:** Re-examine the “either/or” policy for the bike benefit and MBTA pass to allow greater flexibility for staff and faculty with complex or hybrid commutes.
3. **Targeted Support for Staff Living Far from Campus:** Consider additional subsidies or support for staff who live more than five miles from campus and are not served by the MBTA, especially those required to be on-site during paid parking hours.
4. **Engage Staff in Policy Development:** Conduct focus groups and surveys with custodial, facilities, and other staff to ensure their needs and perspectives directly inform parking and commuter benefit reforms.

Addressing faculty and staff transportation needs through a sliding scale parking policy, more flexible commuter benefits, and targeted support for those with limited alternatives will advance equity and well-being across the MIT community.

3 Data Collection

A robust, multi-method data collection strategy grounds this project in the lived experiences and needs of MIT’s diverse community. The following approaches have been and will continue to be pursued:

1. Focus Groups and Stakeholder Interviews

- **Low-Income and Off-Campus Students:** Conducted focus groups to identify transportation barriers, especially regarding the cost of leaving campus, grocery access, and perceptions of biking and transit safety.
- **Program Staff and Advising:** Initiated talks with program staff from MIT FLI Advising, ARM Coalition, and the Office of the First Year to gain qualitative insights on student transportation challenges. S³ referred the project to Naomi Lane, who is collaborating on graduate student shuttle needs.
- **Graduate Student Community:** Communicate with Naomi Lane and graduate student representatives to understand shuttle and transit service gaps.

2. Campus-Wide Surveys

- **Biking Survey ($n > 180$):** Assessed comfort levels, barriers, and interest in biking and MIT PE biking classes. Key findings include high demand for street safety education and a significant portion of students who do not know how to ride a bike or are uncomfortable biking in city traffic.
- **Transportation Habits and Barriers Survey ($n > 160$):** Quantified the prevalence of cost-related trip avoidance, long walks with groceries, and destinations students wish were more accessible. Collected open-ended testimonials to capture the real impact of transportation barriers.

3. Institutional Data and Existing Reports

- **Commuting to MIT Surveys (2021, 2023):** Analyzed to supplement primary data and benchmark commuting patterns, modal splits, and staff parking needs.
- **Topographical Mapping:** Used to illustrate the physical challenges of biking to and from campus (e.g., uphill routes to Harvard, bridge to Boston).

4. Ongoing and Planned Institutional Engagement

- **MITOS and Bluebikes:** MITOS must initiate formal discussions with Bluebikes management to address e-bike allocation, subsidy expansion, and targeted communications.

- **MIT Parking & Transportation Office:** Continue collaboration to monitor progress on the M2 shuttle partnership and to explore sliding scale parking policy and flexible commuter benefits.
- **MIT Physical Education:** Continue working with MIT PE to develop and potentially launch biking classes and workshops tailored to MIT’s unique urban context.
- **Graduate Student and Shuttle Services:** Continue dialogue with Naomi Lane and graduate student leaders to ensure shuttle solutions meet the needs of both undergraduate and graduate populations.

5. Qualitative Testimonies

- Integrated dozens of direct student and staff quotes from surveys and focus groups to illustrate the real-world impact of transportation barriers on daily life, food security, and well-being.

6. Assigning Costs and Data to Each Solution

- **Cost-Benefit Analysis:** MITOS must work with relevant offices to assign estimated costs, potential funding sources, and projected impact for each proposed solution (e.g., Bluebikes subsidy expansion, targeted MBTA passes, sliding scale parking).
- **Data-Driven Advocacy:** Use survey data, focus group findings, and institutional benchmarks to demonstrate both the need and the expected benefits of each intervention. This includes quantifying the number of students and staff affected, the frequency of cost-related barriers, and the potential for increased sustainable mode share.
- **Equity and Accessibility Metrics:** Incorporate metrics such as reduced trip avoidance, increased grocery access, improved biking confidence, and greater staff satisfaction to make the case for each recommendation.

Summary

This data-driven approach ensures that all recommendations are grounded in the real needs and priorities of MIT’s most impacted populations. As the project moves forward, continued partnership with MITOS, Bluebikes, MIT Parking & Transportation, MIT PE, and student/staff groups will be critical for refining solutions and advocating for effective, equitable change.